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Ankamah-Kusi et al.

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(54) **ELECTRONIC DEVICE MULTILEVEL
PACKAGE SUBSTRATE FOR IMPROVED
ELECTROMIGRATION PERFORMANCE**

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(56) **References Cited**

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U.S. PATENT DOCUMENTS

7,989,944 B2 * 8/2011 Tuominen H05K 1/185
257/E23.005
8,129,226 B2 * 3/2012 Johnston H01L 23/3128
438/106
8,922,005 B2 * 12/2014 Hu H01L 25/105
257/737
9,107,290 B1 * 8/2015 Chen H05K 3/0061
(Continued)

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OTHER PUBLICATIONS

PCT Search Report PCT/US2002/040587 mailed Jan. 19, 2023, 2
pages.

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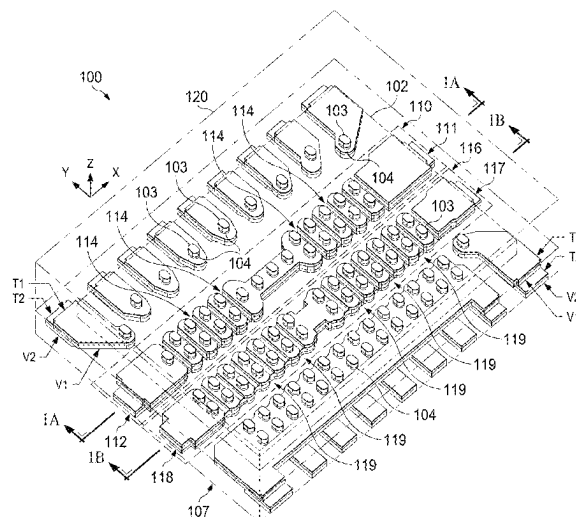
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(57) **ABSTRACT**

An electronic device includes a multilevel package substrate
with first and second levels extending in planes of first and
second directions and spaced apart from one another along
a third direction, the first level having a first side with
landing areas spaced apart from one another along the first
direction. The multilevel package substrate includes a con-
ductive structure having first and second ends and conduc-
tive portions in the first and second levels that provide a
conductive path along the first direction from the landing
areas toward the second end, where the conductive structure
includes indents that extend into the conductive portions in
the first level, the indents spaced apart from one another
along the first direction and positioned along the first direc-
tion between respective pairs of the landing areas.

18 Claims, 11 Drawing Sheets



(51)	Int. Cl.	(56)	References Cited
	<i>H01L 21/683</i> (2006.01) <i>H01L 23/00</i> (2006.01)		U.S. PATENT DOCUMENTS
(52)	U.S. Cl.		
	CPC <i>H01L 21/6835</i> (2013.01); <i>H01L 24/14</i> (2013.01); <i>H01L 24/16</i> (2013.01); <i>H01L 24/17</i> (2013.01); <i>H01L 24/81</i> (2013.01); <i>H01L 2221/68345</i> (2013.01); <i>H01L 2224/1416</i> (2013.01); <i>H01L 2224/16225</i> (2013.01); <i>H01L 2224/17106</i> (2013.01); <i>H01L 2224/81385</i> (2013.01); <i>H01L 2224/81815</i> (2013.01)		9,257,386 B2 * 2/2016 Arisaka H01L 23/49822 9,466,631 B2 10/2016 Fallica et al. 10,068,874 B2 * 9/2018 Nelson H01L 25/0655 10,153,177 B2 * 12/2018 Sorimachi H01L 23/49838 10,262,932 B2 * 4/2019 Furuichi H01L 21/6835 10,905,005 B2 * 1/2021 Taneda H05K 1/181 11,417,613 B2 * 8/2022 Kim H01L 23/5386 11,978,699 B2 * 5/2024 Ankamah-Kusi H01L 21/561 2010/0212947 A1 8/2010 Yamanaka et al. 2016/0066423 A1 3/2016 Sakamoto et al. 2020/0066716 A1 2/2020 Tang et al. 2020/0211971 A1 7/2020 Mizutani 2020/0411416 A1 12/2020 Koduri et al. 2021/0143103 A1 * 5/2021 Mrad H01L 23/49833 2021/0175195 A1 6/2021 Chen et al.
(58)	Field of Classification Search		
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	USPC 257/737		
	See application file for complete search history.		* cited by examiner

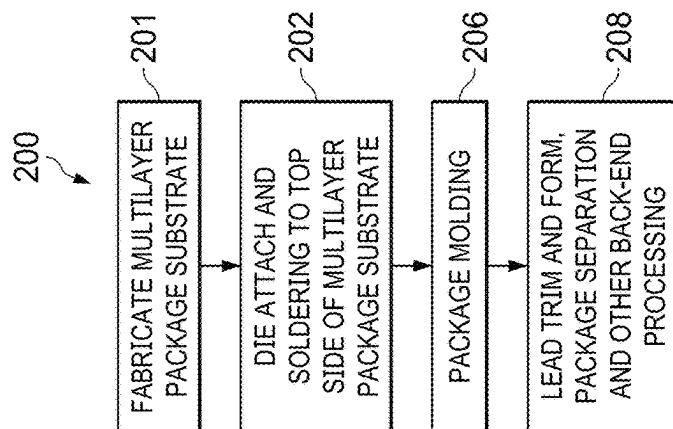
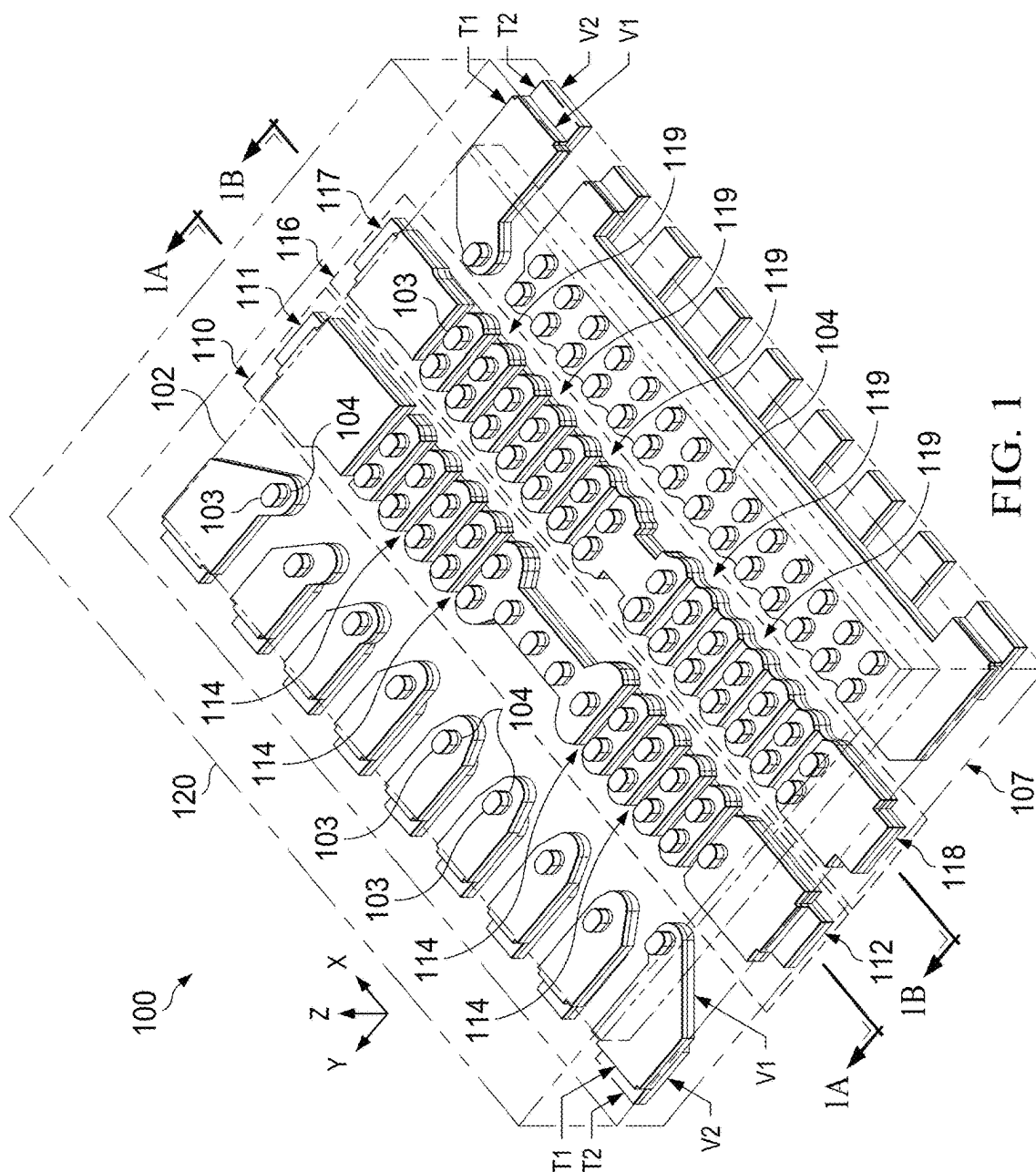
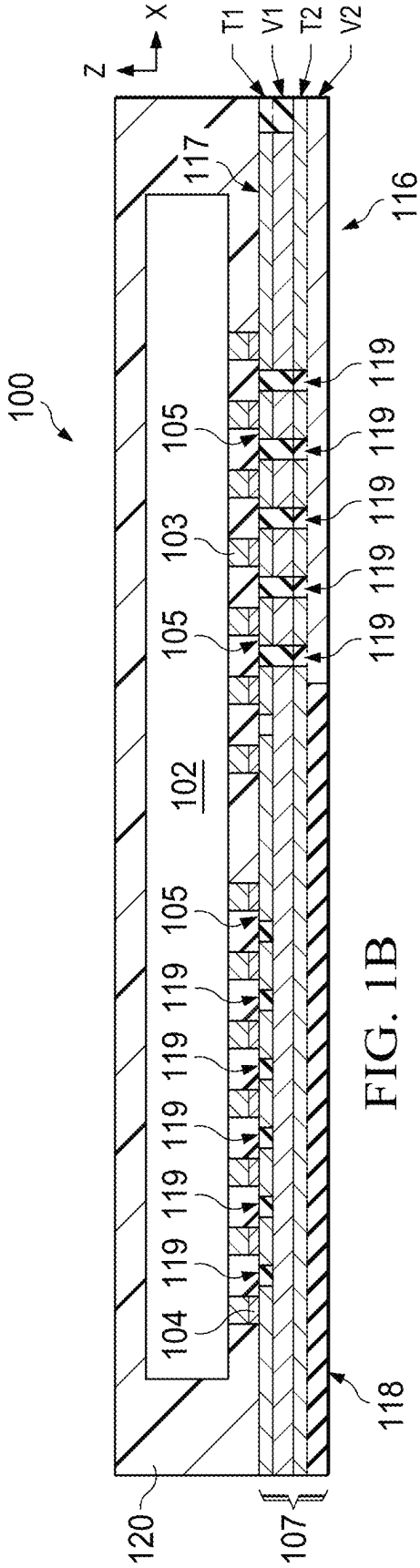
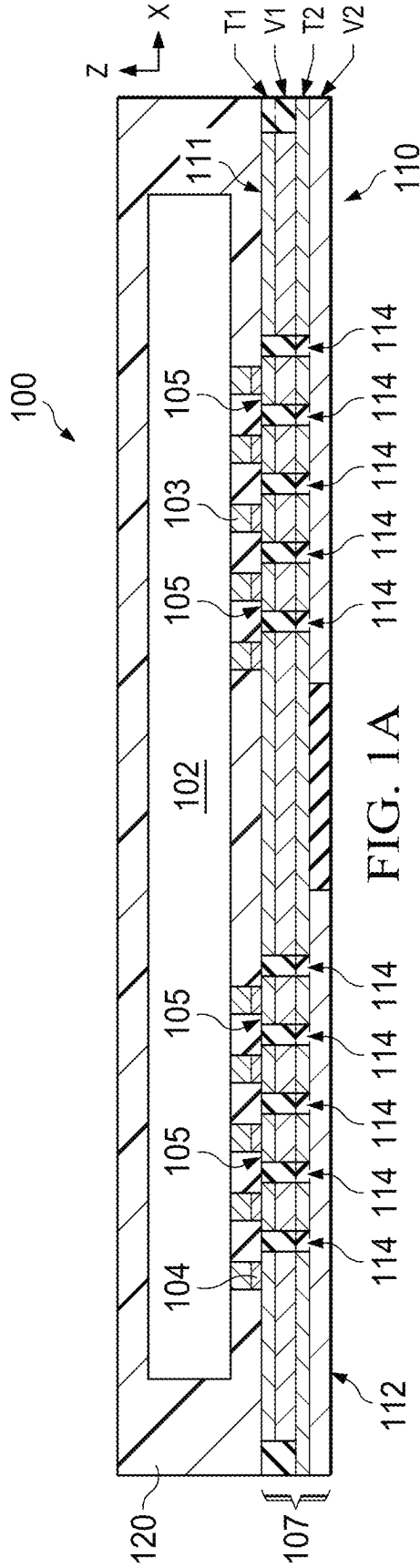
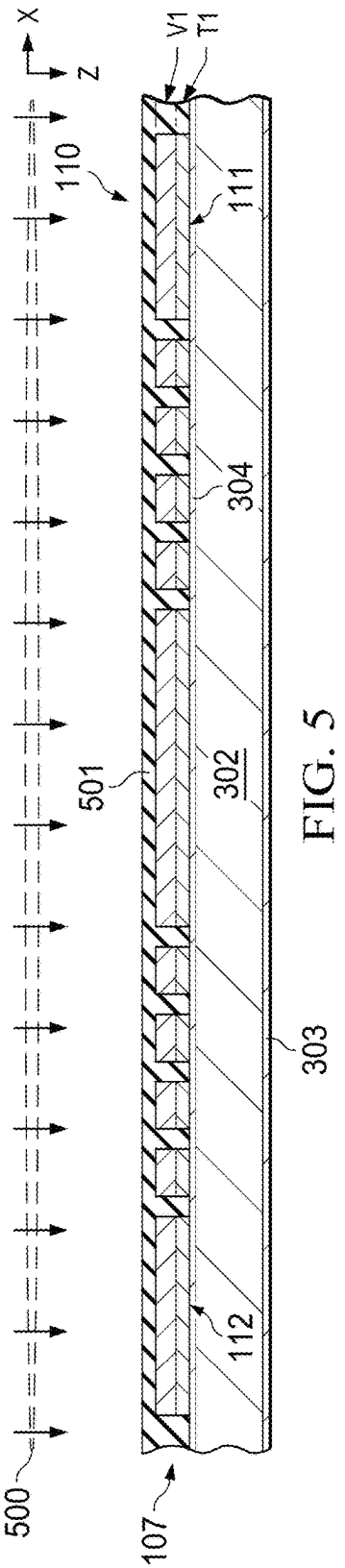
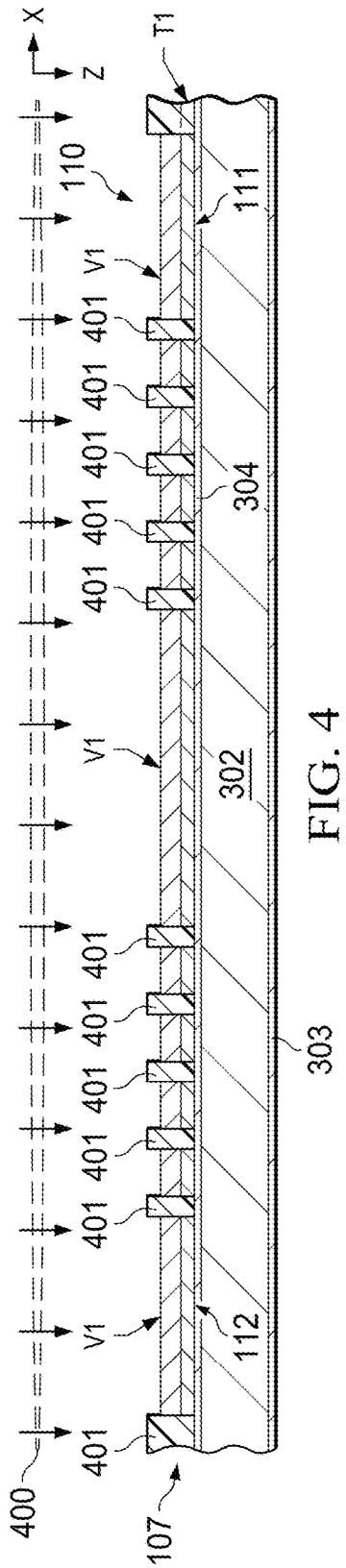
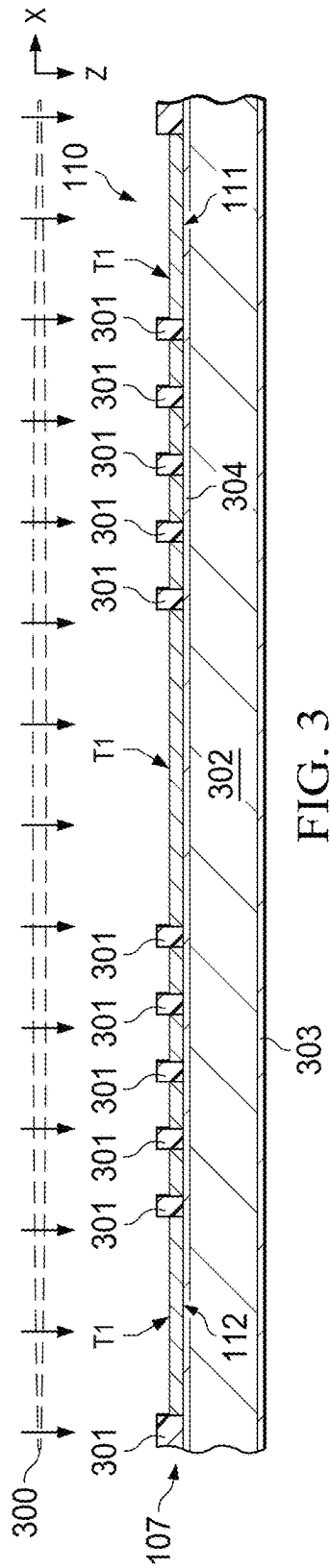
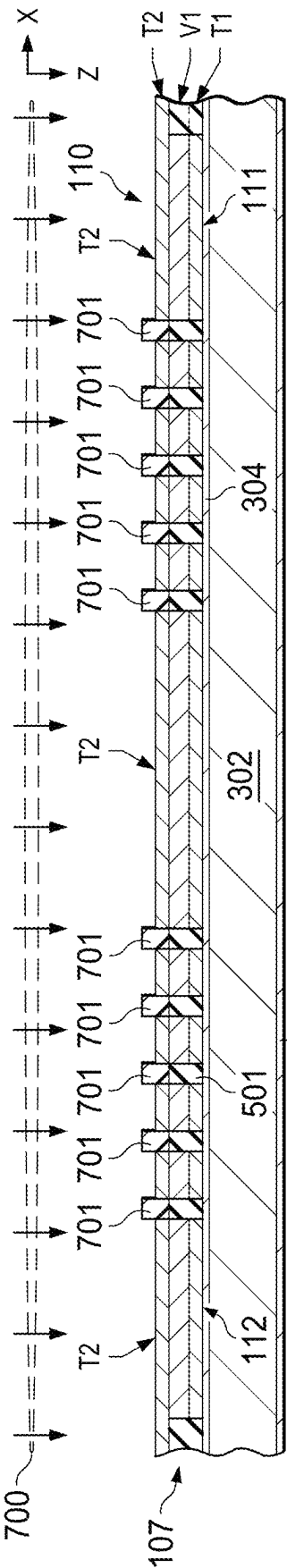
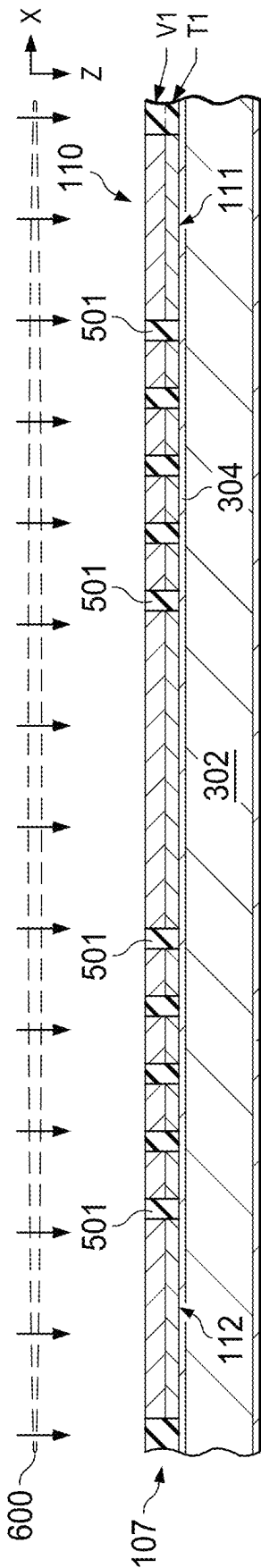


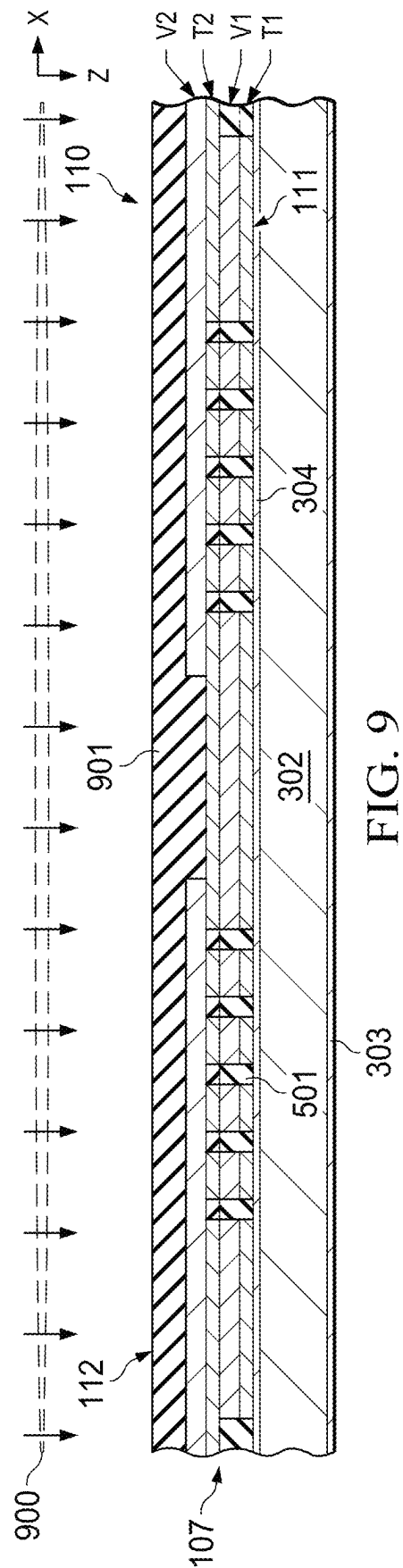
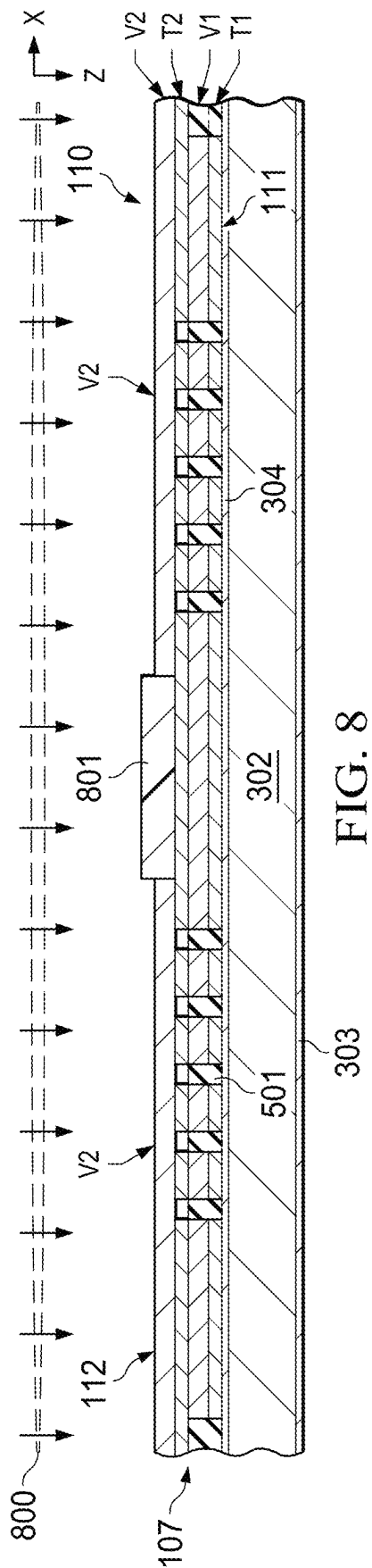
FIG. 2

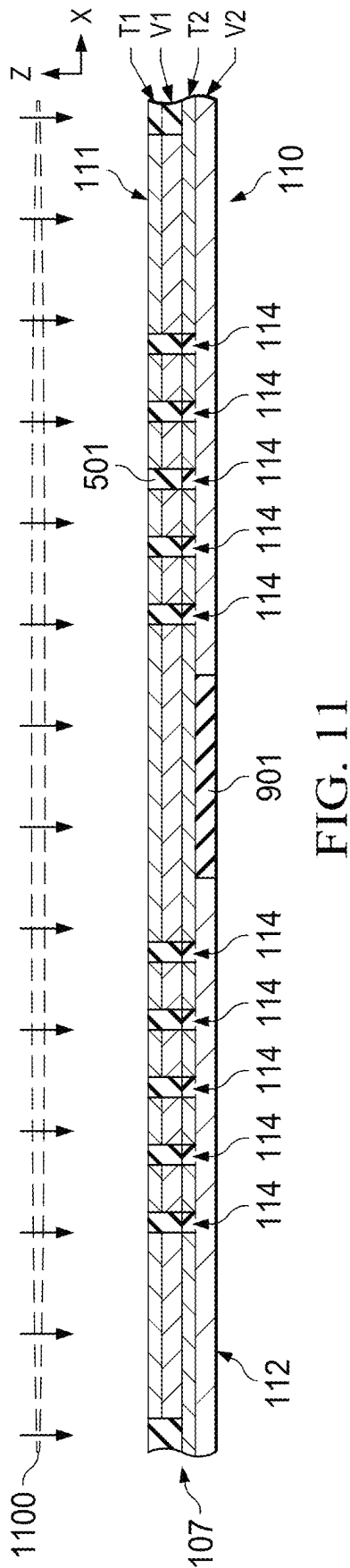
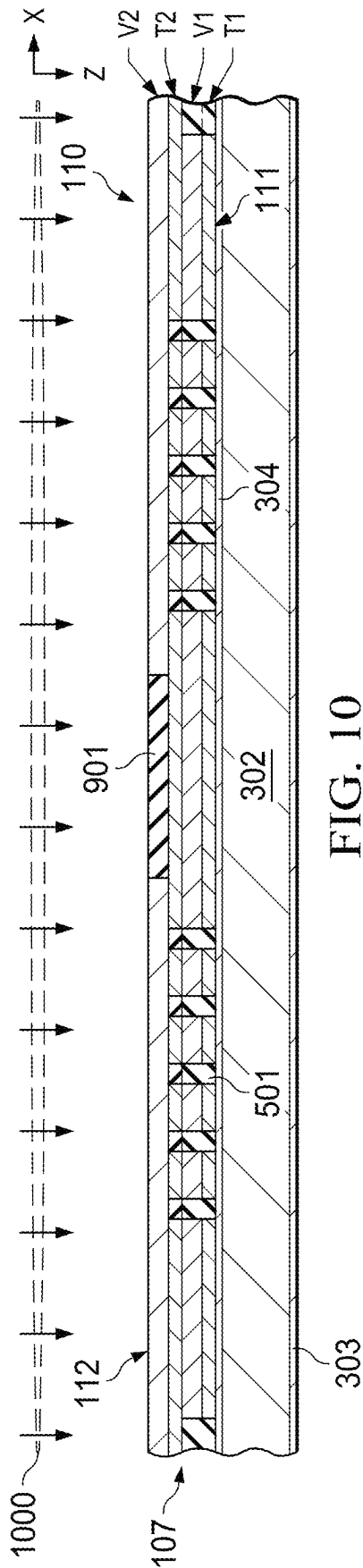
FIG. 1











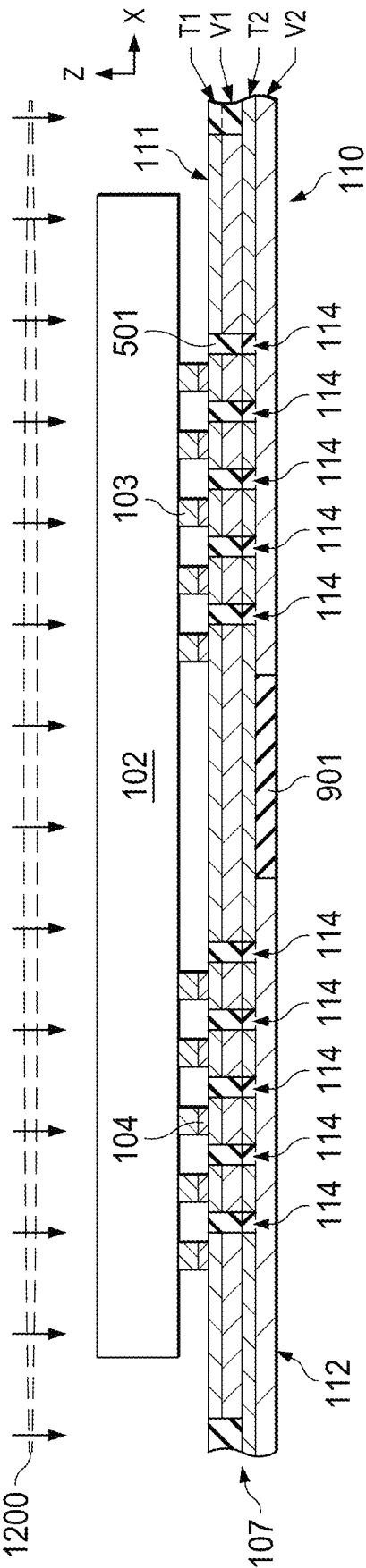


FIG. 12

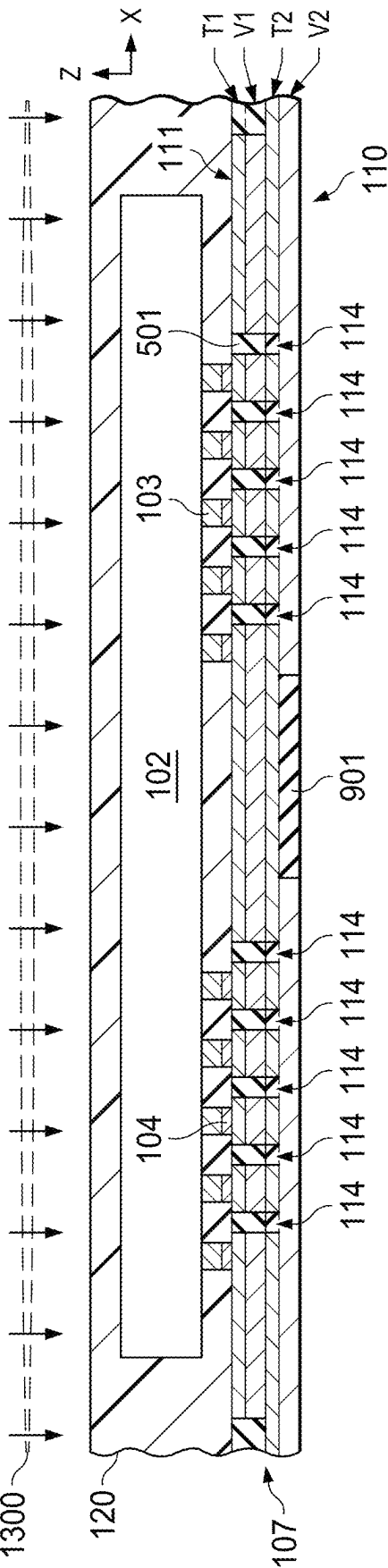
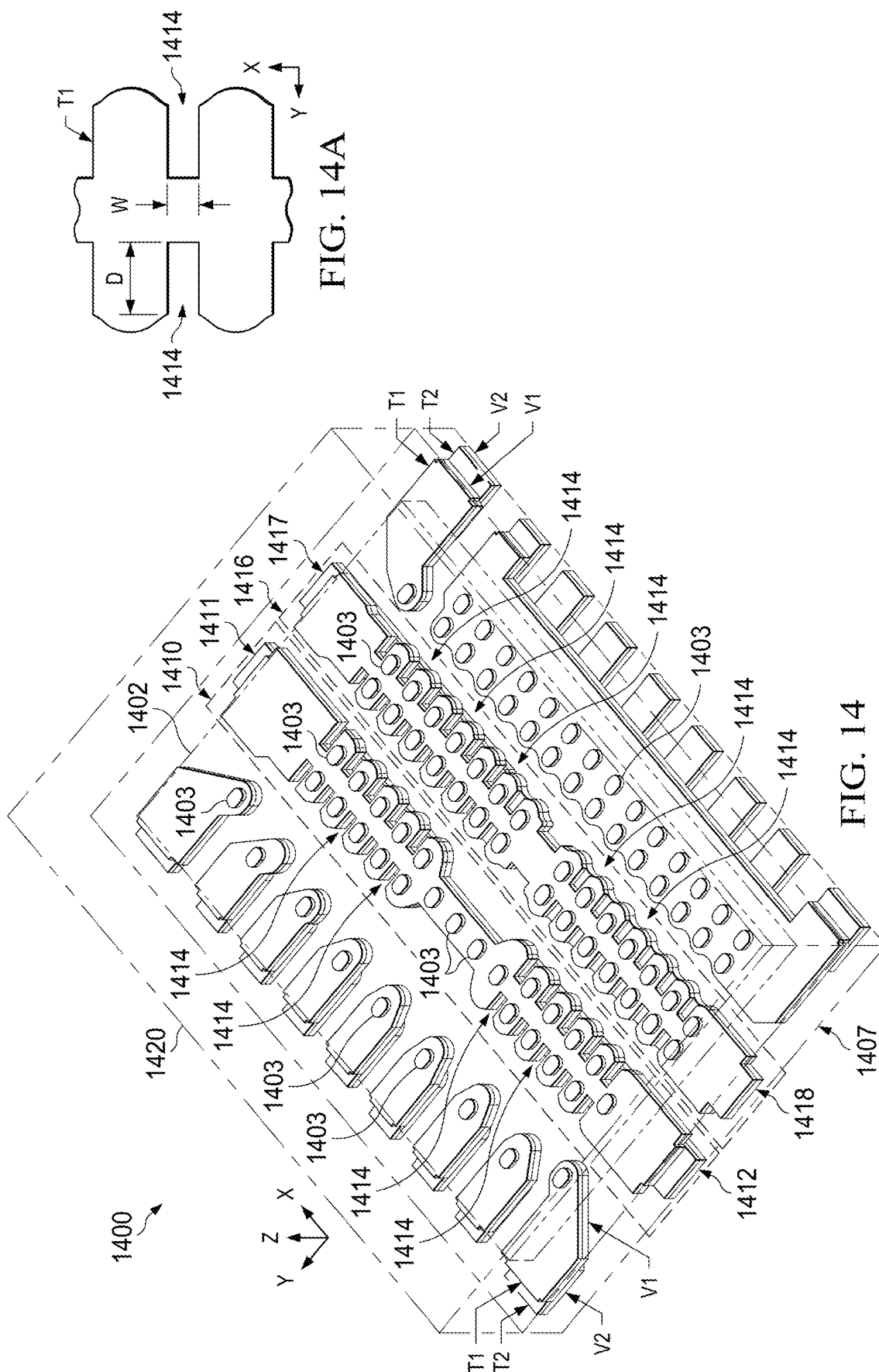


FIG. 13



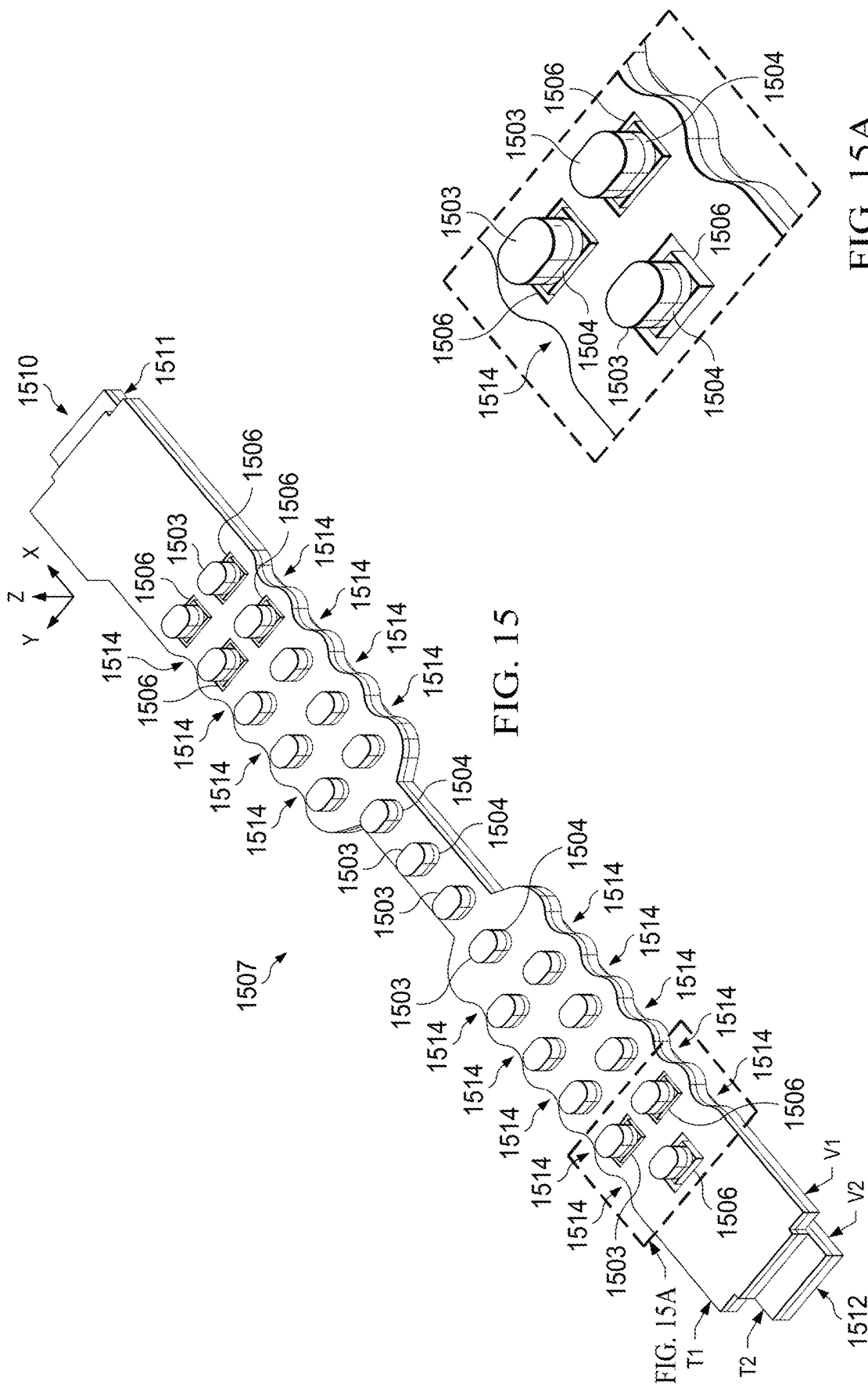


FIG. 15A

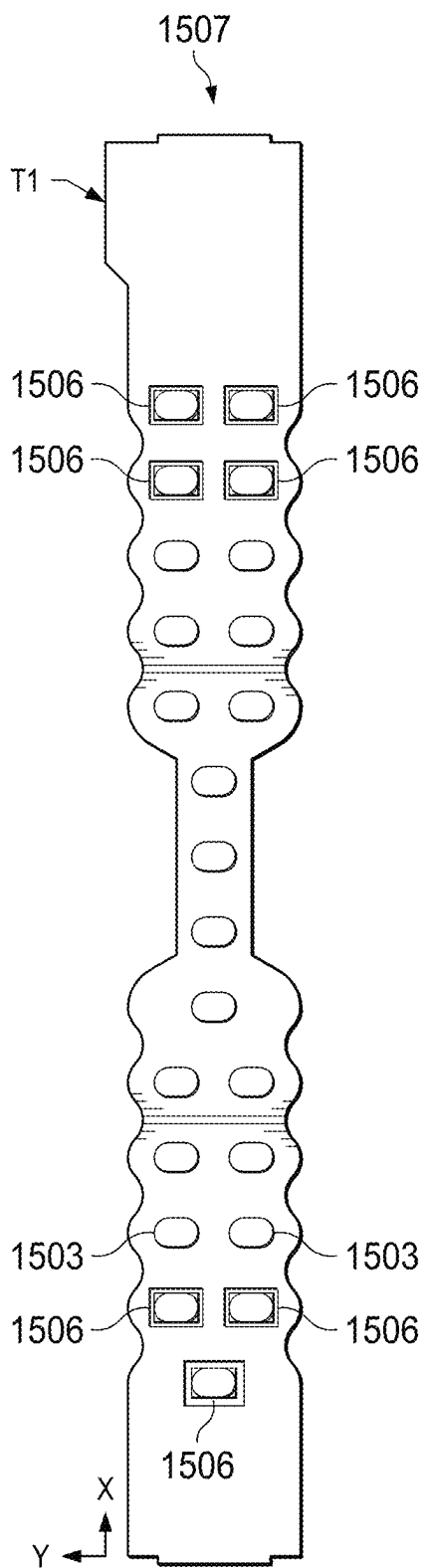


FIG. 15B

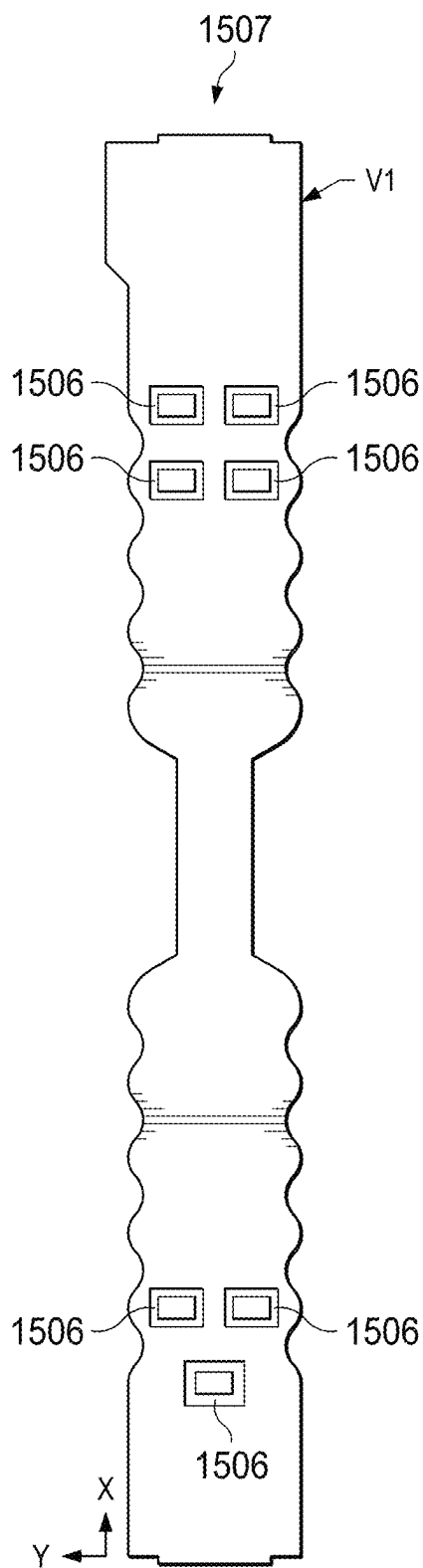


FIG. 15C

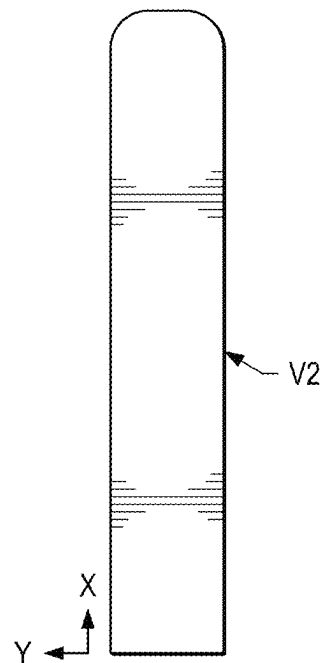
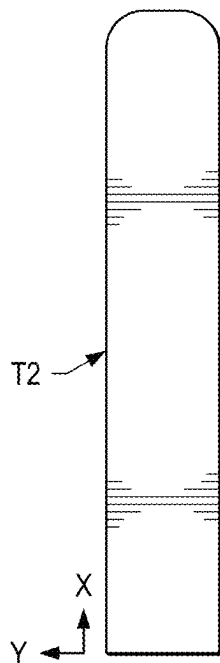
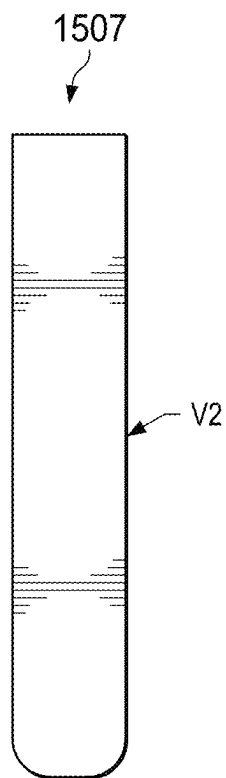
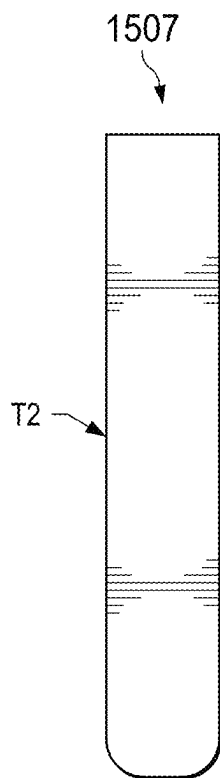


FIG. 15D

FIG. 15E

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ELECTRONIC DEVICE MULTILEVEL PACKAGE SUBSTRATE FOR IMPROVED ELECTROMIGRATION PERFORMANCE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of application Ser. No. 17/406,150 filed Aug. 19, 2021, now U.S. Pat. No. 11,978,699, which is incorporated by reference herein in its entirety.

BACKGROUND

As the size of packaged electronic devices is reduced, power and current density increases. Higher current carrying capability can be improved by using large power posts, such as conductive bumps or pillars on a die, which are soldered to a lead frame. However, electromigration (EM) can cause adverse effects in high current packaged electronic devices where multiple die pillars or bumps are soldered to a lead frame. Coarse line spacing package design rules can lead to current crowding, high current density, and thermal hot spots at the solder/bump interface of certain die pillars or bumps. The high currents and temperatures of the solder interfaces can cause electromigration and lead to premature device performance degradation.

SUMMARY

In one aspect, an electronic device includes a multilevel package substrate, a die, and a package structure that encloses the die and a portion of the multilevel package substrate. The multilevel package substrate has a first level, a second level, a conductive structure, and conductive leads. The individual first and second levels have patterned conductive features and molded dielectric features. The first level extends in a plane of a first direction and an orthogonal second direction and includes a conductive first trace layer and a conductive first via layer. The first level has a first side with landing areas spaced apart from one another along the first direction. The die has conductive terminals electrically coupled to respective ones of the landing areas. The second level includes a conductive second trace layer and a conductive second via layer. The second trace layer is spaced apart from the first trace layer along a third direction that is orthogonal to the first and second directions. The conductive structure has a first end and a second end that is spaced apart from the first end along the first direction. The conductive structure includes conductive portions in the first and second levels that provide a conductive path along the first direction from the landing areas toward the second end. The conductive structure includes indents that extend into the conductive portions in the first level. The indents are spaced apart from one another along the first direction, and at least some of the indents are positioned along the first direction between respective pairs of the landing areas.

In another aspect, a multilevel package substrate includes first and second levels with patterned conductive features and molded dielectric features. The first level extends in a plane of a first direction and an orthogonal second direction and includes a conductive first trace layer and a conductive first via layer. The first level has a first side with landing areas spaced apart from one another along the first direction. The second level includes a conductive second trace layer and a conductive second via layer. The second trace layer is spaced apart from the first trace layer along a third direction that is orthogonal to the first and second directions. The

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multilevel package substrate includes a conductive structure having a first end and a second end that is spaced apart from the first end along the first direction. The conductive structure includes conductive portions in the first and second levels that provide a conductive path along the first direction from the landing areas toward the second end. The conductive structure includes indents that extend into the conductive portions in the first level. The indents are spaced apart from one another along the first direction, and the indents are positioned along the first direction between respective pairs of the landing areas.

In a further aspect, a method for fabricating an electronic device includes fabricating a multilevel package substrate by forming a first level on a carrier structure. The first level has first patterned conductive features and first molded dielectric features and extends in a plane of a first direction and an orthogonal second direction, the first level includes a conductive first trace layer and a conductive first via layer. The first level has a first side with landing areas spaced apart from one another along the first direction. The method also includes forming a second level on the first level. The second level has second patterned conductive features and second molded dielectric features. The second level includes a conductive second trace layer and a conductive second via layer. The second trace layer is spaced apart from the first trace layer along a third direction that is orthogonal to the first and second directions. The method further includes forming a conductive structure in the first and second levels. The conductive structure has a first end and a second end that is spaced apart from the first end along the first direction. The conductive structure includes conductive portions in the first and second levels that provide a conductive path along the first direction from the landing areas toward the second end. The conductive structure includes indents that extend into the conductive portions in the first level. The indents are spaced apart from one another along the first direction, and the indents are positioned along the first direction between respective pairs of the landing areas. The method also includes removing the carrier structure from the first level, performing an electrical connection process that electrically couples conductive terminals of a die to respective ones of the landing areas, and performing a molding process that encloses the die and a portion of the multilevel package substrate in a package structure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a top perspective view of a packaged electronic device that includes a multilevel package substrate with a conductive structure having top side slots in a multilevel package substrate.

FIG. 1A is a partial sectional side elevation view of the electronic device taken along line 1A-1A in FIG. 1.

FIG. 1B is a partial sectional side elevation view of the electronic device taken along line 1B-1B in FIG. 1.

FIG. 2 is a flow diagram of a method for fabricating an electronic device.

FIGS. 3-13 are partial sectional side elevation views of the electronic device of FIG. 1 undergoing fabrication according to the method for FIG. 2.

FIG. 14 is a top perspective view of another packaged electronic device having a conductive structure with lateral indents in a multilevel package substrate.

FIG. 14A is a partial top view of a portion of the multilevel package substrate of FIG. 14 showing example lateral indents.

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FIG. 15 is a top perspective view of another conductive structure with top side moats in a multilevel package substrate.

FIG. 15A is a partial top perspective view of a portion of the multilevel package substrate of FIG. 15 showing example top side moats.

FIG. 15B is a top plan view of a first trace layer of the multilevel package substrate of FIG. 15 showing example top side moats.

FIG. 15C is a top plan view of a first via layer of the multilevel package substrate of FIG. 15 showing example top side moats.

FIG. 15D is a top plan view of a second trace layer of the multilevel package substrate of FIG. 15.

FIG. 15E is a top plan view of a second via layer of the multilevel package substrate of FIG. 15.

DETAILED DESCRIPTION

In the drawings, like reference numerals refer to like elements throughout, and the various features are not necessarily drawn to scale. Also, the term “couple” or “couples” includes indirect or direct electrical or mechanical connection or combinations thereof. For example, if a first device couples to or is coupled with a second device, that connection may be through a direct electrical connection, or through an indirect electrical connection via one or more intervening devices and connections. One or more operational characteristics of various circuits, systems and/or components are hereinafter described in the context of functions which in some cases result from configuration and/or interconnection of various structures when circuitry is powered and operating.

Referring initially to FIGS. 1, 1A, and 1B, a packaged electronic device 100 includes a semiconductor die with conductive bumps or pillars soldered to landing areas of a multilevel package substrate. The multilevel package substrate includes a conductive structure with indents to mitigate electromigration by controlling current flowing to improve device performance and extend the device mean time to failure (MTTF). The electronic device 100 facilitates reduced electromigration effects by mitigating current crowding while allowing direct soldering of the conductive bumps or pillars of the die to the multilevel package substrate landing areas. This provides a low-cost solution compared to other approaches that use barrier metals such as nickel (Ni), nickel tungsten (NiW), etc. to avoid direct contact between copper (Cu) and tin (Sn). Another approach is to use tapered or pedestal lead frame features for solder connections to increase the area of the solder interface, but this approach also increases manufacturing cost and complexity compared to the described examples. The described examples facilitate the reduction in device size and cost while accommodating enhanced power and current density in packaged electronic devices and mitigating current crowding and thermal hot spots in conductive (e.g., copper) die posts and the interface with the solder at the connection to the multilevel package substrate.

Increased current density and resulting current and thermal hotspots can decrease the device mean time to failure per the following equation: $MTTF = A/J^n \exp(E_a/KT)$, where A is a geometry constant of the structure, J is the average current density, n is an empirically determined current density exponent (e.g., 2 to 3), E_a is the activation energy, K is the Boltzmann constant, and T is the device temperature. In one implementation, the landing areas of the multilevel package substrate in the electronic device are or

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include electroplated copper. The conductive features of the multilevel package substrate are designed with indented features to mitigate current crowding and electromigration in the die conductive terminals and the solder interface of the die terminals electrically coupled to respective ones of the landing areas.

FIG. 1 shows a top view of the packaged electronic device 100, FIG. 1A shows a sectional side view of the electronic device 100 taken along line 1A-1A in FIG. 1, and FIG. 1B shows a sectional side view taken along line 1B-1B in FIG. 1. The electronic device 100 includes a semiconductor die 102 having electronic components therein, such as transistors, resistors, capacitors, etc. that are interconnected in a circuit, with conductive terminals 103 electrically coupled by solder connections 104 to respective landing areas 105 (e.g., FIGS. 1A and 1B) of a multilevel package substrate 107. The multilevel package substrate 107 has a first level that includes a first trace layer T1 and a first via layer V1, as well as a second level that includes a second trace layer T2 and a second via layer V2. In other examples, the multilevel package substrate includes more than two levels. The first and second levels T1, V1 and T2, V2 each have patterned conductive features, such as copper, aluminum, or other conductive metal, as well as compression molded dielectric features between different conductive features and between adjacent levels. The molded dielectric features in one example are or include an electrically insulating dielectric material, where the thickness and material in the respective levels provide a withstanding voltage according to a desired voltage separation between circuits and components thereof for a given design.

The multilevel package substrate 107 also has conductive structures 110 and 116, as well as conductive leads with exposed lower surfaces to facilitate soldering to a host printed circuit board (PCB, not shown). The first level T1, V1 extends in a plane of a first direction (e.g., labeled X in the drawings) and an orthogonal second direction (e.g., labeled Y). The first level T1, V1 includes a conductive first trace layer T1 and a conductive first via layer V1. The first level T1, V1 has a first side (e.g., the top side of the first trace layer T1) with the landing areas 105. The landing areas 105 are spaced apart from one another along the first direction X. The second level T2, V2 includes a conductive second trace layer T2 and a conductive second via layer V2. The second trace layer T2 is spaced apart from (e.g., below) the first trace layer T1 along a third direction (e.g., labeled Z) that is orthogonal to the first and second directions X and Y.

In this example, the first conductive structure 110 has a first end 111 and a second end 112 that is spaced apart from the first end 111 along the first direction X. The second conductive structure 116 has a first end 117 and a second end 118 that is spaced apart from the first end 117 along the first direction X. The conductive structures 110 and 116 are generally straight and provide respective conductive paths generally parallel to the first direction X. In other examples, curved or curvilinear or piecewise linear shapes are possible. The individual conductive structures 110 and 116 include conductive portions in the first level T1, V1 and in the second level T2, V2 that provide a conductive path along the first direction X from the landing areas 105 toward the respective second ends 112 and 118.

The electronic device 100 also includes a package structure 120 that encloses the die 102 and a portion of the multilevel package substrate 107. In one example, the package structure 120 is or includes a molded material, such as plastic. In another example, the package structure 120 is or includes a ceramic material.

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The first conductive structure **110** includes indents **114** that extend into the conductive portions in the first level **T1**, **V1**. The second conductive structure **116** in this example includes indents **119** that extend into the conductive portions in the first level **T1**, **V1**. The indents in this example are grooves or trenches that extend downward along the third direction **Z** (e.g., in the negative **Z** direction). In addition, the first and second conductive structures **110** and **116** in this example further include second indents that extend into the conductive structures **110** and **116** along the second direction **Y**, to provide lateral saw tooth shaped sidewalls as shown in FIG. 1.

As best shown in FIGS. 1A and 1B, some of the indents **114** and **119** in this example further extend into the conductive features in the second level **T2**, **V2**. For example, the groove indents **114** and **119** nearest to the respective first ends **111** and **117** extend through the first trace layer **T1**, the first via layer **V1**, and into or through the second trace layer **T2**. However, the groove indents **119** of the second conductive structure **116** nearest to the second end **118** extend through the first trace layer **T1** and the first via layer **V1**, but do not extend through or into the second trace layer **T2** or the second via layer **V2**. Any combination of groove indent depths (e.g., downward along the third direction **Z**) can be used in other implementations, and different indents **114**, **119** can have different groove indent depths, and can be of any suitable profile, having straight portions, curved portions, or combinations thereof. The lateral (e.g., saw tooth) indents can be of different lengths (e.g., inward along the second direction **Y**), and can be of any suitable profile, having straight portions, curved portions, or combinations thereof.

The groove indents **114** and **119** are spaced apart from one another along the first direction **X**. In addition, the indents **114** and **119** are positioned along the first direction **X** between respective pairs of the landing areas **105**. This facilitates controlled current densities in the conductive structures **110** and **116** to mitigate current crowding and thermal hot spots in the conductive structures **110** and **116** and in the conductive terminals **103** and their interfaces with the solder connections **104** to the respective landing areas **105** of the multilevel package substrate **107**. The conductive structures **110** and **116** in one example provide plated copper interconnect structures that facilitate much finer design rules compared to lead frame structures, and different implementations can be customized for different electromigration control applications. FIGS. 14-15E below show further example implementations using different indent features and combinations thereof.

FIG. 2 shows a method **200** for fabricating an electronic device, and FIGS. 3-13 show sectional side views of the electronic device **100** taken along line 1A-1A of FIG. 1 undergoing fabrication according to the method **200**. At **201**, the method **200** includes fabricating the multilevel package substrate **107**. In one example, the multilevel package substrate fabrication at **201** includes forming the first level (e.g., **T1**, **V1**) on a carrier structure, and forming the second level (e.g., **T2**, **V2**) on the first level, after which the carrier structure is removed from the first level. The formation of the first and second levels in this example, concurrently forms the conductive structures **110** and **116** thereof, which are defined in the illustrated implementation by the plating masks used in electroplating operations detailed below.

FIGS. 3-6 show formation of the first level **T1**, **V1**, for example, using an electroplating process **300** and a patterned plating mask **301**. The illustrated example forms the first level having the first molded dielectric features, the first

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patterned conductive features, and the conductive leads of the eventual electronic device **100**. The first level formation starts with forming the first trace layer **T1** using a stainless-steel carrier **302**, such as a panel or strip with multiple prospective package substrate sections, one of which is shown in FIG. 3. The carrier structure **302** includes thin copper seed layers **303** and **304** on the respective bottom and top sides of the carrier structure **302** to facilitate the electroplating process **300**. The electroplating process **300** deposits copper onto the upper seed layer **304** in the portions of the top side of the carrier structure that are exposed through the patterned plating mask **301**.

FIG. 4 shows the multilevel package substrate **107** after the process **300** is completed and the plating mask **301** has been removed. A second electroplating process **400** is performed in FIG. 4 using a patterned second plating mask **401** (e.g., a copper pillar plating process). The electroplating process **400** deposits further copper to form the first via layer **V1** in the areas exposed by the second plating mask **401**. After the process **400** is completed, the second plating mask **401** is removed.

FIGS. 5 and 6 show the formation of the first molded dielectric features in the first level. A compression molding process **500** is performed in FIG. 5 that forms molded dielectric features **501** on exposed portions of the conductive features of the first trace layer **T1** and the vias of the first via layer **V1** to an initial thickness that covers the first trace layer **T1** and the first via layer **V1**. A grinding process **600** is performed in FIG. 6, which grinds upper portions of the molded dielectric material **501** and exposes the upper portions of the first trace layer **T1** and the first via layer **V1**. In another example, a chemical etch is used. In a further example, a chemical mechanical polishing process is used.

FIGS. 7-10 show formation of the second level of the multilevel package substrate **107**, including forming the second trace layer **T2**, the second via layer **V2** and the second molded dielectric features. In one example, the processing used to form the second level is similar to that used to form the first level, although not a requirement of all possible implementations. In the illustrated example, the second level processing forms the second level **T2**, **V2** on the first level **T1**, **V1**, where the second level **T2**, **V2** has the second patterned conductive features and second molded dielectric features. FIG. 7 shows the multilevel package substrate **107** undergoing an electroplating process **700** with a patterned plating mask **701**. The electroplating process **700** deposits copper onto the top side of the portions of the finished first level that are exposed through the plating mask **701** to form the second trace layer. After the process **700** is completed, the plating mask **701** is removed. FIG. 8 shows the multilevel package substrate **107** undergoing another electroplating process **800** using another plating mask **801** (e.g., a copper pillar plating process). The electroplating process **800** deposits further copper to form the second via layer **V2** in the areas exposed by the plating mask **801**. After the process **800** is completed, the plating mask **801** is removed.

FIGS. 9 and 10 show formation of the second molded dielectric features in the second level using compression molding and grinding. A compression molding process **900** is performed in FIG. 9, which forms molded dielectric features **901** on exposed portions of the second via layer **V2** to an initial thickness that covers the second trace layer **T2** and the second via layer **V2**. A grinding process **1000** is performed in FIG. 10, which grinds upper portions of the molded dielectric material **901** and exposes the upper portions of the second trace layer **T2** and the second via layer

V2. In another example, a chemical etch is used. In a further example, a chemical mechanical polishing process is used.

A process 1100 is performed in FIG. 11 that removes the carrier structure 302, 303, 304 from the first level T1, V1. This leaves the finished multilevel package substrate 107 as shown in FIG. 11. The method 200 in FIG. 2 also includes die attach and soldering at 202 via an electrical connection process 1200 in FIG. 12 that electrically couples the conductive terminals 103 of the semiconductor die 102 to respective ones of the landing areas 105 on the top side of the first trace layer T1. In FIG. 12, solder 104 is applied (e.g., dipped or otherwise deposited) onto the bottom sides of the conductive terminals 103 of the semiconductor die 102, and the semiconductor die 102 is placed with the respective terminals 103 on or over the respective landing areas 105 of the first trace layer T1. A thermal solder reflow process 1200 is performed that solders the die terminals 103 to the landing areas 105.

The method 200 continues at 206 in FIG. 2 with package molding. FIG. 13 shows one example, in which a molding process 1300 is performed that encloses a portion of the multilevel package substrate 107 and the die 102 in the package structure 120. A package separation process (not shown) is then performed at 208 in FIG. 2 (e.g., sawing, laser cutting, etc.) that separates individual packaged electronic devices 100 from a panel of concurrently processed devices, where FIG. 1 above shows the example finished electronic device 100.

In one example, the dielectric layers of the multilevel package substrate 107 are or include MJ1 ABF RLF dielectric material, and the package structure 120 is or includes Carsem/TITL mold compound. In one example, all or portions of the device leads are or include copper formed as described above as part of the multilevel package substrate fabrication processing. In another example, all or portions of the conductive leads are or include solder balls. In another example, the multilevel package substrate 107 is mounted to a die attach pad of a lead frame (now shown), which is then molded to provide a finished packaged electronic device.

FIG. 14 shows another packaged electronic device 1400 having conductive structures with lateral indents in a multilevel package substrate. The electronic device 1400 in this example has conductive structures 1410 and 1416 with lateral (e.g., saw tooth) indents 1414 that extend into the lateral sides of the conductive structures 1410 and 1416 along the second direction Y. FIG. 14A shows a top view of a portion of the multilevel package substrate of FIG. 14 with example lateral indent width dimensions W (e.g., 0.05 μm) and depth dimensions D (e.g., 0.11 μm). The electronic device 1400 in this example includes a semiconductor die 1402 having electronic components therein, such as transistors, resistors, capacitors, etc. that are interconnected in a circuit, with conductive terminals 1403 electrically coupled by solder connections (not numerically shown in FIG. 14, e.g., 104 in FIGS. 1-1B above) to respective landing areas (not numerically shown in FIG. 14, e.g., 105 in FIGS. 1A and 1B above) of a multilevel package substrate 1407.

The multilevel package substrate 1407 has a first level that includes a first trace layer T1 and a first via layer V1, as well as a second level that includes a second trace layer T2 and a second via layer V2. In other examples, the multilevel package substrate includes more than two levels. The first and second levels T1, V1 and T2, V2 each have patterned conductive features, such as copper, aluminum, or other conductive metal, as well as compression molded dielectric features between different conductive features and between adjacent levels. The molded dielectric features in one

example are or include an electrically insulating dielectric material, where the thickness and material in the respective levels provide a withstanding voltage according to a desired voltage separation between circuits and components thereof for a given design.

The multilevel package substrate 1407 also includes the respective first and second conductive structures 1410 and 1416, as well as conductive leads with exposed lower surfaces to facilitate soldering to a host printed circuit board (PCB, not shown). The first level T1, V1 extends in a plane of a first direction (e.g., labeled X in FIGS. 14 and 14A) and an orthogonal second direction (e.g., labeled Y). The first level T1, V1 includes a conductive first trace layer T1 and a conductive first via layer V1. The first level T1, V1 has a first side (e.g., the top side of the first trace layer T1) with the landing areas. The landing areas are spaced apart from one another along the first direction X. The second level T2, V2 includes a conductive second trace layer T2 and a conductive second via layer V2. The second trace layer T2 is spaced apart from (e.g., below) the first trace layer T1 along a third direction (e.g., labeled Z) that is orthogonal to the first and second directions X and Y.

The first conductive structure 1410 has a first end 1411 and a second end 1412 that is spaced apart from the first end 1411 along the first direction X. The second conductive structure 1416 has a first end 1417 and a second end 1418 that is spaced apart from the first end 1417 along the first direction X. The conductive structures 1410 and 1416 are generally straight and provide respective conductive paths generally parallel to the first direction X. In other examples, curved or curvilinear or piecewise linear shapes are possible. The individual conductive structures 1410 and 1416 include conductive portions in the first level T1, V1 and in the second level T2, V2 that provide a conductive path along the first direction X from the landing areas toward the respective second ends 1412 and 1418. The electronic device 1400 also includes a package structure 1420 that encloses the die 1402 and a portion of the multilevel package substrate 1407. In one example, the package structure 1420 is or includes a molded material, such as plastic. In another example, the package structure 1420 is or includes a ceramic material.

The first conductive structure 1410 includes indents 1414 that extend into the conductive portions in the first level T1, V1. The second conductive structure 1416 in this example also includes indents 1414. The indents 1414 extend into the conductive portions in the first level T1, V1. The indents 1414 in this example extend laterally into the conductive structures 1410 and 1416 along the second direction Y, to provide lateral saw tooth shaped sidewalls as shown in FIG. 14. The indents 1414 in this example further extend into the conductive features in the second level T2, V2. For example, the indents 1414 extend through the first trace layer T1, the first via layer V1, and the second trace layer T2, and some extend into and through the second via layer V2. Any combination of groove indent depths (e.g., downward along the third direction Z) can be used in other implementations, and different indents 1414 can have different groove indent depths, and can be of any suitable profile, having straight portions, curved portions, or combinations thereof. In addition, the lateral (e.g., saw tooth) indents 1414 can be of different lengths (e.g., inward along the second direction Y), and can be of any suitable profile, having straight portions, curved portions, or combinations thereof.

The saw tooth indents 1414 are spaced apart from one another along the first direction X. In addition, at least some of the indents 1414 are positioned along the first direction X between respective pairs of the landing areas. This facilitates

controlled current densities in the conductive structures **1410** and **1416** to mitigate current crowding and thermal hot spots in the conductive structures **1410** and **1416** and in the conductive terminals **1403** and their interfaces with the solder connections to the respective landing areas of the multilevel package substrate **1407**. The conductive structures in one example provide plated copper interconnect structures that facilitate much finer design rules compared to lead frame structures, and different implementations can be customized for different electromigration control applications.

FIGS. **15-15E** show a conductive structure portion **1510** in a multilevel package substrate **1507** of another example electronic device. This example includes top side moat indents **1506** in the conductive structure **1510**. The moat indents **1506** create an island in the upper first level **T1**, **V1** that includes an associated landing area to limit the current flow through select landing areas to better balance the current densities within the conductive structure **1510**. The conductive structure **1510** in this example also has lateral (e.g., saw tooth) second indents **1514** that extend into the lateral sides of the conductive structure **1510** along the second direction **Y**. In another implementation, the lateral indents **1514** are omitted. FIG. **15A** shows a portion of the multilevel package substrate **1507** with example top side moat indents **1506**. FIG. **15B** shows a first trace layer of the conductive structure **1510** with example top side moat indents **1506**. FIG. **15C** shows a first via layer of the conductive structure **1510** with example top side moat indents **1506**. FIG. **15D** shows a second trace layer of the conductive structure **1510** and FIG. **15E** shows a second via layer of the conductive structure **1510**.

The moat indents **1506** extend into the conductive structure **1510** along the third direction **Z**. In this example, the indents **1506** and **1514** extend through the first trace layer **T1** along the third direction **Z**, and the moat indents **1514** laterally surround respective landing areas of the first trace layer **T1**. In the illustrated example, moat indents **1506** are provided around certain ones of the landing areas nearest to the ends **1511** and **1512**, while interior landing areas of the conductive structure **1510** do not have moat indents (e.g., FIGS. **15B** and **15C**). In another example, more or fewer of the landing areas have associated moat indents **1506**. Moreover, as shown in FIGS. **15B** and **15C**, different moat indents **1506** can have different lateral dimensions in an **X-Y** plane of the first and second directions. Furthermore, different moat indents **1506** can have different shapes/profiles in various implementations.

The conductive structure portion **1510** as shown is part of a multilevel package substrate **1507** to which a semiconductor die (not shown) is soldered, with conductive terminals **1503** of the die electrically coupled by solder connections **1504** to respective landing areas (not numerically shown in FIG. **15**, e.g., **105** in FIGS. **1A** and **1B** above) of the multilevel package substrate **1507**. The multilevel package substrate **1507** has a first level that includes the first trace layer **T1** and a first via layer **V1**, as well as a second level that includes a second trace layer **T2** and a second via layer **V2**. In other examples, the multilevel package substrate includes more than two levels. The first and second levels **T1**, **V1** and **T2**, **V2** each have patterned conductive features, such as copper, aluminum, or other conductive metal, as well as compression molded dielectric features between different conductive features and between adjacent levels. The molded dielectric features in one example are or include an electrically insulating dielectric material, where the thickness and material in the respective levels provide a with-

standing voltage according to a desired voltage separation between circuits and components thereof for a given design.

The multilevel package substrate **1507** also includes the conductive structure **1510** as well as conductive leads (not shown) with exposed lower surfaces to facilitate soldering to a host printed circuit board (PCB, now shown). The first level **T1**, **V1** extends in a plane of a first direction (e.g., labeled **X** in FIG. **15**) and an orthogonal second direction (e.g., labeled **Y**). The first level **T1**, **V1** includes the conductive first trace layer **T1** and a conductive first via layer **V1**. The first level **T1**, **V1** has a first side (e.g., the top side of the first trace layer **T1**) with the landing areas. The landing areas are spaced apart from one another along the first direction **X**. The second level **T2**, **V2** includes a conductive second trace layer **T2** and a conductive second via layer **V2**. The second trace layer **T2** is spaced apart from (e.g., below) the first trace layer **T1** along a third direction (e.g., labeled **Z**) that is orthogonal to the first and second directions **X** and **Y**.

The conductive structure **1510** has a first end **1511** and a second end **1512** that is spaced apart from the first end **1511** along the first direction **X**. The conductive structure **1510** is generally straight and provides a conductive path generally parallel to the first direction **X**. In other examples, curved or curvilinear or piecewise linear shapes are possible. The conductive structure **1510** includes conductive portions in the first level **T1**, **V1** and in the second level **T2**, **V2** that provide a conductive path along the first direction **X** from the landing areas toward the second end **1512**. The electronic device **1500** also includes a package structure (not shown, e.g., **120** in FIG. **1** above) that encloses the die and a portion of the multilevel package substrate **1507**. In one example, the package structure is or includes a molded material, such as plastic. In another example, the package structure is or includes a ceramic material.

As further shown in FIGS. **15B-15E**, the moat indents **1506** extend downward along the third direction **Z** into the conductive portions in the first level **T1**, **V1** (e.g., FIGS. **15B** and **15C**), but do not extend into the second level **T2**, **V2** (FIGS. **15D** and **15E**). The lateral second indents **1514** extend along the second direction **Y** into the conductive portions in the first level **T1**, **V1** (e.g., FIGS. **15B** and **15C**) to provide partially lateral saw tooth shaped upper sidewalls, but do not extend into the second level **T2**, **V2** (FIGS. **15D** and **15E**). Any combination of moat indent depths and saw tooth indent depths (e.g., downward along the third direction **Z**) can be used in other implementations, and different indents **1506** and/or **1514** can have different lateral dimensions and shapes, and the indents **1506** and **1514** can be of any suitable profile (e.g., having straight portions, curved portions, or combinations thereof).

The moat indents **1506** and saw tooth indents **1514** in this example are spaced apart from one another along the first direction **X**. In addition, at least some of the indents **1514** are positioned along the first direction **X** between respective pairs of the landing areas. The moat indents **1506**, whether alone or in combination with the lateral saw tooth indents **1514** help control current densities to enhance current density uniformity and mitigate crowding and hot spots in the conductive structure **1510** as well as in the conductive terminals **1503** and their interfaces with the solder connections to the respective landing areas of the multilevel package substrate **1507**. The illustrated examples facilitate current density uniformity and reduced electromigration to extend product device lifetime and combat performance degradation, while supporting higher power and current density as electronic device package sizes are reduced. For

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transistor device examples, the described examples facilitate balanced, uniform current density while maintaining acceptable transistor current flow with little or no adverse impact on device on-state resistance (e.g., Rdson).

Modifications are possible in the described examples, and other implementations are possible, within the scope of the claims.

What is claimed is:

1. A method for making an electronic device, comprising: forming a multilevel package substrate having a first level, a second level, a conductive structure, and conductive leads; forming molded dielectric features and patterning conductive features in the first and second levels; extending the first level in a plane of a first direction and an orthogonal second direction, the first level including a conductive first trace layer and a conductive first via layer, the first level having a first side with landing areas spaced apart from one another along the first direction; the second level including a conductive second trace layer and a conductive second via layer, the second trace layer spaced apart from the first trace layer along a third direction that is orthogonal to the first and second directions; the conductive structure having a first end and a second end that is spaced apart from the first end along the first direction, the conductive structure including conductive portions in the first and second levels that provide a conductive path along the first direction from the landing areas toward the second end; forming indents in the conductive structure that extend into the conductive portions in the first level, the indents spaced apart from one another along the first direction, and the indents positioned along the first direction between respective pairs of the landing areas; electrically coupling conductive terminals of a die to respective ones of the landing areas; and enclosing the die and a portion of the multilevel package substrate with a package structure.
2. The method for claim 1, wherein the indents extend into the conductive features in the second level.
3. The method for claim 2, wherein the indents extend into the conductive structure along the third direction.
4. The method for claim 3, wherein the indents extend through the conductive portions in the first level.
5. The method for claim 3, wherein: the indents extend through the first trace layer along a third direction; and the indents laterally surround the landing areas.
6. The method for claim 3, further including extending second indents into the conductive structure along the second direction.
7. The method for claim 2, wherein the indents extend into the conductive structure along the second direction.

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8. The electronic device of claim 1, wherein the indents extend into the conductive structure along the third direction.

9. The electronic device of claim 8, wherein the indents extend through the conductive portions in the first level.

10. The method for claim 8, wherein: the indents extend through the first trace layer along a third direction; and the indents laterally surround the landing areas.

11. The method for claim 1, wherein the indents extend into the conductive structure along the second direction.

12. A method for making a multilevel package substrate, comprising:

- forming a first level;
- forming a second level; and
- forming a conductive structure;
- the first and second levels each having patterned conductive features and molded dielectric features;
- the first level extending in a plane of a first direction and an orthogonal second direction, the first level including a conductive first trace layer and a conductive first via layer, the first level having a first side with landing areas spaced apart from one another along the first direction, the landing areas adapted to be soldered to conductive terminals of a die;
- the second level including a conductive second trace layer and a conductive second via layer, the second trace layer spaced apart from the first trace layer along a third direction that is orthogonal to the first and second directions;
- the conductive structure having a first end and a second end that is spaced apart from the first end along the first direction, the conductive structure including conductive portions in the first and second levels that provide a conductive path along the first direction from the landing areas toward the second end; and
- extending indents from the conductive structure into the conductive portions in the first level, the indents spaced apart from one another along the first direction, and the indents positioned along the first direction between respective pairs of the landing areas.
13. The method for claim 12, wherein the indents extend into the conductive features in the second level.
14. The method for claim 13, wherein the indents extend into the conductive structure along the second direction.
15. The method for claim 12, wherein the indents extend into the conductive structure along the third direction.
16. The method for claim 15, wherein the indents extend through the conductive portions in the first level.
17. The method for claim 15, wherein: the indents extend through the first trace layer along a third direction; and the indents laterally surround the landing areas.
18. The method for multilevel package substrate of claim 12, wherein the indents extend into the conductive structure along the second direction.

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